

NAVJOT SINGH CHAHAL

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5+ years of experience in systems software, high-performance computing, and distributed machine learning inference. Proven expertise in C++, CUDA, custom kernel authoring, and robotics libraries.

Education

University of Pennsylvania, Philadelphia, USA **June 2026**

Master of Science (Electrical Engineering at Department of Electrical Systems Engineering), GPA: 3.59/4.0

Kurukshetra University, Kurukshetra, India

June 2019

Bachelor of Technology (Electronics and Communications Engineering), GPA: 4.0/4.0

Professional Experience

University of Pennsylvania

Robotics Research Engineer at Figueroa GRASP Lab

Jan 2026 - Present

- Saved \$30,000 in hardware and eliminated the need for external tactile skins by engineering a virtual skin manifold for contact detection, achieving 0.2 mm precision at 100 Hz using particle filtering and internal joint torques in morphologies.
- Learning object shape in voxels to infer grasping and manipulation strategies for unknown objects in blind state.

Research Assistant at Quantum Hardware Lab

Aug 2025 - Dec 2025

- Engineered an automated quantum control software pipeline for a room-temperature liquid-state NMR system, programmatically sweeping radio-frequency (RF) pulses to map **Rabi chevrons** on the nuclear spins of a water molecule.

UHG Optum, Software Engineer - CTO Acceleration Team, India

March 2022 - Jul 2024

- Accomplished a 35% reduction in storage overhead compared to naive file backups by implementing a content-addressable storage system using SHA-256 hashing. 3x faster parallel file hashing Using Scala Parallel Collections and Future abstractions to process multi-gigabyte staging areas.
- Scala back-end utilizing Akka Actors for multithreading, transforming 8-11 TB of data streams concurrently daily and Implementing the Either monad for error handling, replacing traditional exceptions with type-safe failure states.

HCL Technologies, Software Engineer, India

Jul 2019 - March 2022

- Improved international cheque review operational time and accuracy by 48% by integrating a transformer-based Human script recognition ML engine into Deutsche Bank's systems. Dedicated SpringBoot microservices for automated document processing batch of 1000's of documents daily thus reducing manual operational effort by 80%.

Projects

Heterogeneous Computing Pipeline for Stream Compression (C++ & Custom Kernels)

SOC 2025

- Optimized CPU resource utilization for edge robotics by offloading stream compression to a custom FPGA kernel, SIMD, Unrolling, Neon intrinsics and pragmas, achieving **135 Mbps per thread** throughput via a C++ OpenCL pipeline deployed on a **Xilinx Ultra96** platform.

Deep Neural Network Compression & Inference Profiling

University of Pennsylvania 2024

- Achieved a 75% reduction in the memory footprint of VGG16 (from 500 MB to 277MB) by implementing sparse matrix techniques, pruning, and quantization, enabling edge deployment while preserving inference accuracy.

Low Light Image Enhancement using Transformer

University of Pennsylvania 2024

- Transformer model using retinex-theory achieved a PSNR of 22.7 at 5 FPS using Gradient Graph Laplacian Regularizer.
- Used ReduceLROnPlateau scheduler, Lion optimizer and Charbonnier loss fast convergence training time by 1.5%.

Autonomous F1/10th Robo Racers Car (Real-time Systems)

University of Pennsylvania X-Lab 2025

- 2nd rank in the University Competition by designing perception and control software for autonomous driving using custom sampling based Model Predictive Control MPPI, and Particle Filtering achieving 50 Hz real-time performance.

Optical Flow and Scene Reconstruction

University of Pennsylvania 2025

- Optimized scene recovery and ego-motion tracking accuracy by 7% by implementing Lucas-Kanade optical flow with RANSAC, integrating transformer-based (LoFTR) feature extraction

AES Encryption decoding using Deep Learning

University of Pennsylvania 2026

- Achieved 90% accuracy in decoding AES encryption keys with less than 100 power traces by training a deep learning model one time on several thousand traces on side-channel power analysis data. By Encoding the power traces into spectrograms, preserving temporal and frequency information, and using a CNN architecture to extract features for key prediction.

Technical Skills

- **Languages:** C, C++, CUDA, Python, Bash, Scala, Java, MATLAB, Docker, Git, Linux
- **Frameworks:** ROS2, PyTorch, TensorFlow, TensorRT, OpenCV, OpenCL, NVIDIA Jetson, Isaac Lab, Mujoco, Rviz
- **Skills:** LiDAR, Depth, IMU, Radar, MPC, SLAM, Localization, Xilinx FPGA, Deep Neural Networks, Image Processing, Optimization, Azure, AWS, Docker